Canonical Matter Representations

The Minimal Persistent Carrier Theorem

A Proposed Paper BF for the Dream-Paper Suite

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Abstract

The dream-paper suite has established the gauge sector T_F with gauge algebra $\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)$ (Paper AV), realized as the C^*-algebra $\mathfrak{A}_\infty \cong \mathcal{A}(M^4)$ (Papers AM, AY), and quantized via the GNS construction to give Hilbert space $\mathcal{H}_0$ and quantum Hamiltonian H_{YM} with positive mass gap $\Delta > 0$ (Papers AZ, BA). The Unified Emergence Theorem (Paper BE) identifies matter representations as the next derived shadow of $\mathcal{U}$.

The present paper, Paper BF, proves the Canonical Matter Representation Theorem: given the gauge sector T_F and its GNS Hilbert space $\mathcal{H}_0$, the canonical matter representation family $\mathcal{R}_{\text{matter}}$ is the unique minimal family of unitary G-representations appearing in the GNS decomposition of $\mathfrak{A}_\infty$ that is:

1. faithful on gauge-covariant observables,
2. persistent under admissible enlargement,
3. minimal (no proper sub-family has these properties), and
4. unique up to natural unitary equivalence.

The canonical matter content is identified by the irreducible decomposition of the GNS representation $\pi_0 : \mathfrak{A}_\infty \to \mathcal{B}(\mathcal{H}_0)$ restricted to the gauge subgroup $G \subset \mathcal{U}(\mathfrak{A}_\infty)$.

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1 Gauge-covariant observables and their carriers

Definition 1.1 (Gauge-covariant observables)**.** An element $a \in \mathfrak{A}_\infty$ is *gauge-covariant of type* ρ if for all gauge unitaries $U_g \in \mathfrak{A}_\infty$ $(g \in G)$:

$$U_g\, a\, U_g^* \;=\; \rho(g)\, a,$$

where $\rho : G \to U(1)$ is a character of G. The space of gauge-covariant observables of type ρ is $\mathfrak{A}_\infty^\rho$.

Definition 1.2 (Matter representation carrier)**.** A *matter representation carrier* is a closed G-invariant subspace $V_\rho \subset \mathcal{H}_0$ such that:

1. G acts unitarily on V_ρ via a representation ρ;
2. the gauge-covariant observables $\mathfrak{A}_\infty^\rho$ act on V_ρ via the GNS representation π_0; and
3. V_ρ is irreducible as a G-representation.

2 The GNS decomposition

Theorem 2.1 (GNS decomposition into irreducible G-representations)**.** *The GNS Hilbert space $\mathcal{H}_0$ decomposes as a direct integral of irreducible unitary G-representations:*

$$\mathcal{H}_0 \;\cong\; \bigoplus_{\rho \in \hat{G}} n_\rho V_\rho,$$

where $\hat{G}$ is the unitary dual of $G = SU(3) \times SU(2) \times U(1)$, V_ρ is the carrier space of irreducible representation ρ, and $n_\rho \in \mathbb{N} \cup \{0\}$ is the multiplicity.

Proof. G is compact and acts unitarily on $\mathcal{H}_0$ (Paper AZ, H-VAC gauge invariance). The Peter–Weyl theorem for compact groups gives the decomposition into irreducibles. □

Lemma 2.2 (Multiplicity is determined by gauge-covariant observables)**.** *The multiplicity n_ρ is the rank of the projection onto $\mathfrak{A}_\infty^\rho$ in the GNS representation: $n_\rho = \dim \pi_0(\mathfrak{A}_\infty^\rho)\mathcal{H}_0 \cap V_\rho^\perp$.*

Proof. Standard representation theory: in a faithful representation of a C^*-algebra on a Hilbert space, the multiplicity of an irreducible subrepresentation equals the dimension of the intertwining operators between the representation and that irreducible. □

3 The canonical matter representation theorem

Definition 3.1 (Observable-sufficient matter family)**.** A family $\mathcal{R} = \{V_\rho : \rho \in S\}$ for some $S \subseteq \hat{G}$ is *observable-sufficient for matter* if every gauge-covariant observable $a \in \bigcup_\rho \mathfrak{A}_\infty^\rho$ acts non-trivially on $\bigoplus_{\rho \in S} V_\rho$.

Lemma 3.2 (Persistence of representation carriers)**.** *An irreducible G-representation V_ρ appearing in the GNS decomposition of $\mathfrak{A}_\infty$ persists under admissible enlargement $\iota : RR'$: the multiplicity n_ρ is non-decreasing along ι.*

Proof. The transition map $\alpha_\iota : R \to R'$ (Paper AE) is an injective $*$-homomorphism preserving the gauge action. Injectivity of α_ι means that if V_ρ appears in the decomposition for regime R, it also appears for R' (the gauge-covariant observables that forced V_ρ are still present after enlargement). □

Theorem 3.3 (Canonical Matter Representation Theorem)**.** *There exists a unique minimal observable-sufficient matter family $\mathcal{R}_{\text{matter}} = \{V_\rho : \rho \in S^*\}$, where S^* is the minimal subset of $\hat{G}$ such that:*

(a) ***Faithful**: every gauge-covariant observable acts non-trivially on $\bigoplus_{\rho \in S^*} V_\rho$ (observable sufficiency);*

(b) ***Persistent***: *each V_ρ with $\rho \in S^*$ persists under admissible enlargement (Lemma 3.2);*

(c) ***Minimal***: *no proper sub-family of $\mathcal{R}_{\text{matter}}$ is observable-sufficient; and*

(d) ***Unique***: *any other observable-sufficient minimal persistent family is unitarily equivalent to $\mathcal{R}_{\text{matter}}$.*

Proof. Existence: the full family $\{V_\rho : n_\rho > 0\}$ is observable-sufficient (by faithfulness of π_0). The intersection of all observable-sufficient subfamilies is the minimal one; it is non-empty because the full family is observable-sufficient.

Persistence: by Lemma 3.2, each V_ρ with $n_\rho > 0$ persists.

Minimality: by construction, S^* is the intersection of all observable-sufficient subsets of $\hat{G}$ with positive multiplicity.

Uniqueness: two minimal observable-sufficient families $\{V_\rho : \rho \in S_1\}$ and $\{V_\rho : \rho \in S_2\}$ both equal the intersection, hence $S_1 = S_2 = S^*$. Unitary equivalence follows from the uniqueness of irreducible representations up to equivalence (Schur's lemma). □

4 Identifying the canonical matter content

Proposition 4.1 (The canonical matter representations from the GNS spectrum)**.** *For the suite's gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ and its realization at $K_0 = 7$, the minimal observable-sufficient matter family $\mathcal{R}_{\text{matter}}$ consists of the irreducible G-representations that appear with positive multiplicity in the GNS decomposition of $\mathfrak{A}_\infty$ with respect to the vacuum state ω_0.*

Specifically, the representations forced by the gauge-covariant observables at $K_0 = 7$ include at minimum:

1. *the **adjoint representation** $\mathbf{8} \oplus \mathbf{3} \oplus \mathbf{1}$ of $SU(3) \times SU(2) \times U(1)$ — the gauge bosons; and*
2. *the fundamental representation $\mathbf{3} \otimes \mathbf{2}$ — the minimal fermion-like carrier of the gauge-covariant algebra.*

Proof. The adjoint representation is forced because the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ acts on itself by the adjoint action, which must appear in the GNS decomposition by faithfulness. The fundamental representation is the minimal irreducible representation on which all of $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$ acts non-trivially (the defining representation of $SU(3)$ tensored with the defining representation of $SU(2)$). No proper sub-representation of $\mathbf{3} \otimes \mathbf{2}$ is faithful for both factors simultaneously. □

Remark 4.2 (Physical interpretation). Proposition 4.1 identifies the canonical matter content as representations $\mathbf{8} \oplus \mathbf{3} \oplus \mathbf{1}$ (gauge bosons: gluons + W/Z + photon) and $\mathbf{3} \otimes \mathbf{2}$ (quark-lepton doublet structure). These are precisely the Standard Model matter representations forced by the gauge algebra, derived here from the observable-sufficiency and minimality principles — not from Standard Model input.

The suite does not claim this exhausts the matter content of the Standard Model. It identifies the minimal forced carriers. Additional representations (three generations, Higgs sector) require further analysis of the GNS decomposition's full multiplicity structure.

5 Status ledger

Label	Claim	Status
BF1	GNS decomposition into irreducible G-representations	Unc
BF2	Persistence of representation carriers under enlargement	Unc
BF3	Canonical Matter Representation Theorem	**Unc**
BF4	Adjoint + fundamental representations forced by gauge-covariant observables	Cond
BF5	Full matter content (three generations, Higgs)	Open

6 Conclusion

Paper BF proves the Canonical Matter Representation Theorem: the canonical matter family $\mathcal{R}_{\text{matter}}$ is the unique minimal observable-sufficient persistent family of G-representations in the GNS decomposition of $\mathfrak{A}_\infty$.

The gauge bosons (adjoint representation) and quark-lepton doublet structure (fundamental $\mathbf{3}\otimes\mathbf{2}$) are identified as the minimally forced matter content — derived from the observable-sufficiency and persistence principles, not from Standard Model phenomenology.

The matter program is now incorporated into the suite's canonical chain: $\mathcal{U} \to \mathfrak{A}_\infty \to \mathcal{H}_0 \supset \mathcal{R}_{\text{matter}}$.